

Geographic Information Systems

Week-9: Assignment-9

- 1 A DEM is a raster representation of a surface?
 - (a) Discrete
 - (b) Continuous**
 - (c) Triangulated
 - (d) Quadtree
- 2 The DEM could be generated through techniques such as photogrammetry, Lidar, InSAR, land surveying, etc.?
 - (a) True**
 - (b) False
- 3 Grid (DEM) cell values can be:
 - (a) Only integer numbers
 - (b) Only real (floating) numbers
 - (c) Only positive integer numbers
 - (d) Both positive and negative integer or real (floating) numbers**
- 4 DEMs are commonly built using data collected using remote sensing techniques, but they may also be built from land surveying:
 - (a) False
 - (b) True**
- 5 There is no universal usage of the terms digital elevation model (DEM), digital terrain model (DTM) and digital surface model (DSM) in scientific literature:
 - (a) True**
 - (b) False
- 6 Most of the terrain data providers (e.g. USGS, ERSDAC, CGIAR, ISRO-NRSA) use the term?
 - (a) DSM
 - (b) DTM
 - (c) DEM**
 - (d) DES
- 7 DEMs can be generated through number of ways, however most preferred method of generating DEMs is:
 - (a) Contours
 - (b) Remote sensing based**
 - (c) Aerial photographs
 - (d) None of the above
- 8 DEMs can also be generated using the following method:
 - (a) Stereo-pairs
 - (b) Thermal infrared data
 - (c) Radar data pairs
 - (d) All of the above**

- 9 Spatial Interpolation techniques:
- (a) Turns raw data into useful information by adding greater informative content and value
 - (b) Reveals patterns, trends, and anomalies that might otherwise be missed
 - (c) Provides a check on human intuition
 - (d) All of the above**
- 10 Functional surfaces can be used to represent:
- (a) Terrestrial surfaces that depict the earth's surface
 - (b) Statistical surfaces that describe demographic
 - (c) Mathematical surfaces that are based on arithmetic expressions
 - (d) All of the above**

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